

INF 2040 - Web Client Technologies

Lab Exercises

Task

- ▶ Problem: Create a basic HTML5 structure

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Basic HTML Structure</title>
</head>
<body>
  <h1>Hello World</h1>
  <p>This is a basic HTML5 page.</p>
</body>
</html>
```

Task

- ▶ Problem: Add a header, footer, and main content using semantic elements

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Semantic Layout</title>
</head>
<body>
  <header>
    <h1>My Website</h1>
  </header>
  <main>
    <p>This is the main content of the website.</p>
  </main>
  <footer>
    <p>&copy; 2024 My Website</p>
  </footer>
</body>
</html>
```

Task

- ▶ Problem: Create a navigation menu using the `<nav>` element

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Navigation Menu</title>
</head>
<body>
  <nav>
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#about">About</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
</body>
</html>
```

Task

- ▶ Problem: Insert an image with alternate text

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Image Example</title>
</head>
<body>
  
</body>
</html>
```

Task

- ▶ Problem: Create a simple table with rows and headers

Solution

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Table Example</title>
</head>
<body>
  <table>
    <thead>
      <tr>
        <th>Item</th>
        <th>Price</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Apple</td>
        <td>$1</td>
      </tr>
      <tr>
        <td>Banana</td>
        <td>$0.5</td>
      </tr>
    </tbody>
  </table>
</body>
</html>
```

Task

- ▶ Problem: Add a video element with controls

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Video Example</title>
</head>
<body>
  <video controls>
    <source src="video.mp4" type="video/mp4">
    Your browser does not support the video tag.
  </video>
</body>
</html>
```

Task

- ▶ Add an audio file with playback controls

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Audio Example</title>
</head>
<body>
  <audio controls>
    <source src="audio.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
  </audio>
</body>
</html>
```

Task

- ▶ Problem: Create a simple form with various input types
 - ▶ First Name
 - ▶ Last Name
 - ▶ Age
 - ▶ Hair Color
- ▶ Make the fields required
- ▶ Include proper validation for the age field

Solution

HTML ▾

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Form Example</title>
</head>
<body>

  <form action="/submit-form" method="POST">
    <!-- Name field (required) -->
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required><br>
    <br/>

    <!-- Email field (required) -->
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required><br>
    <br/>

    <!-- Age field (required, with min and max validation) -->
    <label for="age">Age:</label>
    <input type="number" id="age" name="age" min="18" max="100" required><br>
    <br/>

    <!-- Hair color picker (optional) -->
    <label for="hairColor">Hair Color:</label>
    <input type="color" id="hairColor" name="hairColor"><br>
    <br/>

    <!-- Submit button -->
    <input type="submit" value="Submit">
  </form>

</body>
</html>
```

Task

- ▶ Problem: Create a list of items (ordered and unordered)

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Lists Example</title>
</head>
<body>
  <h2>Unordered List</h2>
  <ul>
    <li>Item 1</li>
    <li>Item 2</li>
    <li>Item 3</li>
  </ul>

  <h2>Ordered List</h2>
  <ol>
    <li>First</li>
    <li>Second</li>
    <li>Third</li>
  </ol>
</body>
</html>
```

Task

- ▶ Problem: Embed a YouTube video using `<iframe>`

Solution

HTML ▾

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Embedded Video Example</title>
</head>
<body>
  <iframe width="560" height="315" src="https://www.youtube.com/embed/dQw4w9WgXcQ"
    frameborder="0" allow="autoplay; encrypted-media" allowfullscreen></iframe>
</body>
</html>
```

Task

- ▶ Problem: Create a data list with predefined suggestions

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Datalist Example</title>
</head>
<body>
  <label for="browser">Choose a browser:</label>
  <input list="browsers" id="browser" name="browser">
  <datalist id="browsers">
    <option value="Chrome">
    <option value="Firefox">
    <option value="Safari">
    <option value="Edge">
  </datalist>
</body>
</html>
```

Task

- Problem: Implement a `<figure>` with an image and caption

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Figure Example</title>
</head>
<body>
  <figure>
    
    <figcaption>A breathtaking view</figcaption>
  </figure>
</body>
</html>
```

Task

- ▶ Problem: Create a simple form validation using the required attribute

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Form Validation</title>
</head>
<body>
  <form>
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required><br>
    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required><br>
    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

Task

- ▶ Problem: Add an SVG image to your page

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>SVG Example</title>
</head>
<body>
  <svg width="100" height="100">
    <circle cx="50" cy="50" r="40" stroke="black" stroke-width="3" fill="red" />
  </svg>
</body>
</html>
```

Task

- ▶ Draw a Pie Chart with an SVG

- ▶ <https://jsbin.com/mifakas/edit?html,output>

- ▶ Draw a Pie Chart with a Canvas

- ▶ <https://jsbin.com/pigumed/1/edit?html,output>

Solution

html

 Copy code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>SVG Example</title>
</head>
<body>
  <svg width="100" height="100">
    <circle cx="50" cy="50" r="40" stroke="black" stroke-width="3" fill="red" />
  </svg>
</body>
</html>
```



Questions?